

Research Publications

1. Singh, H.K., Verma, A., Gupta, S.K., Singh, D., & Suman, D.V.P. (2025). *Machine learning-driven optimization of concrete mixes incorporating treated sludge*. **Asian Journal of Civil Engineering**. <https://doi.org/10.1007/s42107-025-01404-w>
2. Verma, A., Singh, H.K., Srivastava, G., & Yadav, P.K. (2025). *Comprehensive groundwater quality assessment in Lucknow, India: A WAWQI and spatial distribution approach*. **Iranian Journal of Science and Technology, Transactions of Civil Engineering**. <https://doi.org/10.1007/s40996-025-01913-3>
3. Mandvi, Patel, P.K., & Singh, H.K. (2024). *Performance analysis of machine learning models for AQI predictions in Gorakhpur City: A critical study*. **Environmental Monitoring and Assessment**. <https://doi.org/10.1007/s10661-024-13107-x>
4. Singh, V.P., & Singh, H.K. (2024). *Assessing groundwater quality at Shivri Municipal Landfill: A combined approach using Water Quality Index, Leachate Pollution Index, and GIS*. **Journal of Indian Association for Environmental Management**.
5. Singh, S., & Singh, H.K. (2024). *Comparative analysis of large-scale composting and home-scale composting in Lucknow, Uttar Pradesh*. **Journal of Indian Association for Environmental Management**.
6. Mandvi, Singh, H.K., Patel, P.K., & Singh, S. (2024). *Evaluation of predictive models for Air Quality Index prediction in an Indian urban area*. **Journal of Indian Association for Environmental Management**.

International/National Conferences:

1. Singh, A.P., Pandey, A., Kumar, A., Chaurasiya, A., Kashyup, R., Gautam, A., Saxena, M., & Singh, H.K. (2023). *Water quality assessment of Gomti River using modelling techniques: A review*. In Proceedings of the International Conference on Frontiers in Desalination, Energy, Environment and Material Sciences for Sustainable Development (FEEMSSD-2023 & InDACon-2023), MMMUT Gorakhpur & KIPM-College of Engineering and Technology, India. DOI: 10.21467/proceedings.161.
2. Singh, H.K., Gupta, S., & Patel, S. (2019). *Quantification, classification, and management of plastic waste*. in Indian Conference on Geotechnical and Geo-Environmental Engineering (ICGGE 2019), MNNIT Allahabad, Prayagraj, India.
3. Patel, S., Gupta, S., & Singh, H.K. (2019). *Composting of food and vegetable waste and its effect on soil properties*. in Indian Conference on Geotechnical and Geo-Environmental Engineering (ICGGE 2019), MNNIT Allahabad, Prayagraj, India.

Book Chapters

1. Patel, P.K., **Singh, H.K.**, & Singh, A. (2025). *Nanomaterials in the degradation of toxic organic pollutants*. In *Nanomaterials in Environmental Remediation*. CRC Press.
2. Patel, P.K., **Singh, H.K.**, & Singh, A. (2025). *Wastewater Remediation and Nanoadsorbents*. In *Nanomaterials in Wastewater Research*. Springer Singapore.